

Iranian Mathematics Competitions

Selected Problems, 1973–2007

Real analysis, combinatorics, inequalities, theory of equations

1973 — First Competition, University of Tehran

Q1-1973-1. Define $f : (0, \infty) \rightarrow \mathbb{R}$ by $f(x) = 0$ if $x \notin \mathbb{Q}$, and $f(p/q) = \frac{1}{p+q}$ for p/q in lowest terms with $p, q \in \mathbb{N}$. (a) Show the limit of f exists at every point of $(0, \infty)$. (b) At which points is f continuous?

Q2-1973-1. For a triangle with sides $a < b < c$, set $S = \max\{\frac{a}{b}, \frac{b}{c}, \frac{c}{a}\} \cdot \min\{\frac{a}{b}, \frac{b}{c}, \frac{c}{a}\}$. Show that $S \geq 1$.

Q3-1973-1. Show that the decimal $0.123456789101112131415\dots$, formed by concatenating the positive integers, is not eventually periodic.

Q4-1973-1. Find two distinct real or complex numbers such that each is the cube of the other.

Q5-1973-1. Let $x, y > 0$ with $x + y = 1$. Prove that $x^x + y^y \geq \sqrt{2}$.

1974 — Second Competition, Shiraz University

Q1-1974-2. For positive reals x, y, z, t and $n \in \mathbb{N}$, prove that

$$\left(\frac{x + y + z + t}{4}\right)^n \leq \frac{x^n + y^n + z^n + t^n}{4}.$$

Q2-1974-2. A particle moves along a line, covering unit distance in unit time, starting and ending at rest. Prove that at some instant the absolute value of its acceleration is at least 4.

Q3-1974-2. In Iran, car plate numbers are five-digit numbers with no zero digit. In big cities, an additional Persian letter (from a 32-letter alphabet) labels the plate. Determine the total number of plates containing the letter “dal”.

1975 — Third Competition, Jondi Shapour University, Ahwaz

Q1-1975-3. Let $f : [a, b] \rightarrow \mathbb{R}$ satisfy $f'' > 0$ on (a, b) . Prove that for every $\xi \in (a, b)$, there exists $x_0 \in [a, b]$ with

$$f'(\xi) = \frac{f(a) - f(x_0)}{a - x_0} \quad \text{or} \quad f'(\xi) = \frac{f(b) - f(x_0)}{b - x_0}.$$

Q2-1975-3. A sequence $(a_n)_{n \geq 1}$ of distinct numbers in $[0, 1]$ is given. (a) Show that in general, no continuous $f : [0, 1] \rightarrow [0, 1]$ with $f(a_n) = a_{n+1}$ need exist. (b) Show that such an f does exist if (a_n) is monotonic.

Q3-1975-3. Without induction, prove

$$\sum_{m=0}^n \left(\frac{n!}{m!(n-m)!} \right)^2 = \frac{(2n)!}{(n!)^2}.$$

1976 — Fourth Competition, University of Tabriz

Q1-1976-4. Let $(\alpha_n)_{n \geq 1}$ be nonnegative reals with $\lim_{n \rightarrow \infty} \alpha_n = 0$. Prove there are infinitely many indices n such that $\alpha_m \leq \alpha_n$ for all $m \geq n$.

Q2-1976-4. Evaluate $\int_0^{\pi/2} \frac{\sin^n x}{\cos^n x + \sin^n x} dx$ for $n \in \mathbb{N}$.

1977 — Fifth Competition, Sharif University of Technology

Q1-1977-5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and assume no value more than twice. Prove that f assumes at least one value exactly once.

Q2-1977-5. For $m, n \in \mathbb{N}$, prove that $\frac{(mn)!}{m!(n!)^m}$ is an integer.

1978 — Sixth Competition, University of Isfahan

Q1-1978-6. If f is continuous on $\mathbb{R}$ and $\int_0^\infty |f(x)| dx < \infty$, prove that

$$\lim_{n \rightarrow \infty} \int_0^\infty \left| f\left(x + \frac{1}{n}\right) - f(x) \right| dx = 0.$$

Q2-1978-6. Let $n_1, \dots, n_k \in \mathbb{Z}$ and let $m_1, \dots, m_k$ be a permutation of them. Prove that $|n_1 - m_1| + \dots + |n_k - m_k|$ is even.

Q3-1978-6. Given 1,000,000 points inside a circle, can one find a straight line avoiding all of them that divides the circle into two parts each containing exactly 500,000 points?

1980 — Seventh Competition, Ferdowsi University of Mashhad

Q1-1980-7. Let $x_n = \frac{n+1}{2^{n+1}} \sum_{k=1}^n \frac{2^k}{k}$ for $n \geq 1$. Prove $\lim_{n \rightarrow \infty} x_n$ exists and find its value.

1984 — Eighth Competition, Shiraz University

Q1-1984-8. Without using derivatives, find the minimum of

$$f(x, y, z) = x^2 + 4y^2 + z^2 - 6x + 4y.$$

1985 — Ninth Competition, Tarbiat Moallem University

Q1-1985-9. Let $f : [a, b] \rightarrow [a, b]$ be continuous and increasing with $f(a) = a$, and set $E = \{x \in [a, b] : f(x) \geq x\}$. Prove $f(E) = E$.

Q2-1985-9. Let $\alpha \in \mathbb{R}$. Prove there is no positive continuous $f : [0, 1] \rightarrow \mathbb{R}$ satisfying

$$\int_0^1 f(x) dx = 1, \quad \int_0^1 xf(x) dx = \alpha, \quad \int_0^1 x^2 f(x) dx = \alpha^2.$$

Q3-1985-9. Find the number of positive integer solutions of $x_1 + x_2 + \cdots + x_m = n$, where $m < n$.

1986 — Tenth Competition, University of Sistan and Baluchestan

Q1-1986-10. Let $f : [1, \infty) \rightarrow \mathbb{R}$ satisfy $f(1) = 1$ and $f'(x) = \frac{1}{x^2 + f(x)^2}$ for $x \geq 1$. Prove $\lim_{x \rightarrow \infty} f(x)$ exists and is less than $1 + \frac{\pi}{4}$.

Q2-1986-10. Prove that $\lim_{n \rightarrow \infty} \int_0^3 \frac{x^2(1-x)x^n}{1+x^{2n}} dx = 0$.

1987 — Eleventh Competition, University of Birjand

Q1-1987-11. Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable, and suppose f and f' have no common zero. Prove that the zero set of f in $[0, 1]$ is finite.

Q2-1987-11. A real number c is given. Show that if one root of $x^3 - \frac{3}{4}x + c = 0$ lies in $[-1, 1]$, then all roots lie in $[-1, 1]$.

Q3-1987-11. Let X have n elements and let $\mathcal{C} \subseteq \mathcal{P}(X)$ satisfy: (i) closed under union, (ii) closed under complement, (iii) $\emptyset \in \mathcal{C}$. Show that $|\mathcal{C}| = 2^k$ for some $k \leq n$.

1988 — Twelfth Competition, Guilan University

Q1-1988-12. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, and suppose there exists $M > 0$ with $|f(x) - f(y)| \geq M|x - y|$ for all x, y . Show that f is one-to-one and surjective.

Q2-1988-12. Let $p > 2$ be prime, and let α, β be two distinct roots of

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^p}{p!} = 0.$$

Show that $\alpha - \beta$, $\alpha + \beta$, and $\alpha\beta$ are irrational.

Q3-1988-12. 1700 people took a 15-question true/false test. No participant answered two consecutive questions correctly, and all answered every question. Must two answer sheets be identical?

Q4-1988-12. Evaluate $1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \cdots$, where $n \in \mathbb{N}$.

1989 — Thirteenth Competition, University of Tehran

Q1-1989-13. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be integrable with $f(x+y) = f(x) + f(y)$ for all x, y . Show that $f(x) = cx$ for some constant c .

Q2-1989-13. Find all polynomials $f(x)$ with rational coefficients such that $f(x)$ is irrational whenever x is irrational.

1990 — Fourteenth Competition, University of Isfahan

Q1-1990-14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable on $\mathbb{R} \setminus \{x_0\}$. If $f' < 0 < f''$ on $x < x_0$ and $f' > 0 > f''$ on $x > x_0$, show that f is not differentiable at x_0 .

Q2-1990-14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, and suppose $[a, b] \subset f([c, d])$. Prove there exists $[r, s] \subseteq [c, d]$ with $f([r, s]) = [a, b]$.

Q3-1990-14. Let $|S| = n$ and let $\mathcal{A} = \{A_1, \dots, A_n\}$ be n distinct subsets of S . Show there exists $x \in S$ such that $A_1 \cup \{x\}, \dots, A_n \cup \{x\}$ are all distinct.

Q4-1990-14. A publisher exhibits 1369 titles, displaying 100 each day, with no pair displayed together on two different days. Determine the maximum number of exhibition days.

Q5-1990-14. Find all nonzero real numbers $a_1, \dots, a_n$ satisfying

$$\sum_{i=1}^n a_i^m = \sum_{i=1}^n a_i \quad \text{for } m = 1, 2, \dots, n+1.$$

1991 — Fifteenth Competition, Ferdowsi University of Mashhad

Q1-1991-15. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be increasing and set $\varphi(x) = \int_0^x f(t) dt$. (a) Show that $\varphi\left(\frac{x+y}{2}\right) \leq \frac{1}{2}(\varphi(x) + \varphi(y))$ for $x, y \geq 0$. (b) Conclude that φ is convex.

1993 — Seventeenth Competition, Shahid Beheshti University

Q1-1993-17. Let f be continuous on $[a, b]$ and differentiable on (a, b) , and suppose its graph is not a line segment. Prove there exists $c \in (a, b)$ with

$$|f'(c)| > \left| \frac{f(b) - f(a)}{b - a} \right|.$$

Q2-1993-17. If $p_1(x), p_2(x), p_3(x), p_4(x)$ are real polynomials, prove that

$$\int_{-1}^x p_1(t)p_3(t) dt \int_{-1}^x p_2(t)p_4(t) dt - \int_{-1}^x p_1(t)p_4(t) dt \int_{-1}^x p_2(t)p_3(t) dt$$

is divisible by $(x+1)^4$.

Q3-1993-17. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous with $f(0) = f(1)$. Prove there exist $0 < a \leq b \leq 1$ with $b - a = \frac{1}{2}$ and $f(b) = f(a)$.

1994 — Eighteenth Competition, Sharif University of Technology

Q1-1994-18. Let f, g be continuous periodic functions of period 1. Show that

$$\lim_{n \rightarrow \infty} \int_0^1 f(x)g(nx) dx = \left(\int_0^1 f(x) dx \right) \left(\int_0^1 g(x) dx \right).$$

1995 — Nineteenth Competition, University of Kerman

Q1-1995-19. Let $f : [0, 1] \rightarrow [0, 1]$ be continuous. Show the equation $2x - \int_0^x f(t) dt = 1$ has exactly one root in $[0, 1]$.

Q2-1995-19. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be continuously differentiable with $\lim_{x \rightarrow \infty} (f(x) + hf'(x)) = 0$ for some $h > 0$. Show that $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} f'(x) = 0$.

1998 — Twenty-Second Competition, University of Ahwaz

Q1-1998-22. Let $f : [0, 1] \rightarrow \mathbb{R}$ be differentiable with $f(0) = 0$ and $f(1) = 1$. Show that for each $n \in \mathbb{N}$, there exist $x_1, \dots, x_n \in [0, 1]$ with

$$\sum_{i=1}^n \frac{1}{f'(x_i)} = n.$$

2000 — Twenty-Fourth Competition, Khajeh Nasir Toosi University

Q1-2000-24. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous with g' existing on $[a, b]$. Show that if $(f(a) - g'(a))(g'(b) - f(b)) > 0$, then there exists $c \in (a, b)$ with $f(c) = g'(c)$.

Q2-2000-24. Let f be continuous on $[a, b]$ with $\int_a^b f(t) dt \neq 0$. Show that for each $k \in (0, 1)$, there exists $c \in (a, b)$ with

$$\int_a^c f(t) dt = k \int_a^b f(t) dt.$$

2001 — Twenty-Fifth Competition, IKIU of Qazvin

Q1-2001-25. Let f be twice differentiable with $f''(t) < 0$ for all $t \in \mathbb{R}$. Prove that if $f'(y) + x < f(y + 1)$ for some reals x, y , then $f(y) > x$.

Q2-2001-25. Let $f : (a, b) \rightarrow \mathbb{R}$ be continuously differentiable with

$$\lim_{x \rightarrow a^+} f(x)^2 = 0 \quad \text{and} \quad \lim_{x \rightarrow b^-} f(x)^2 = e - 1.$$

Prove that if $2f(x)f'(x) - f(x)^2 \geq 1$ on (a, b) , then $0 < b - a \leq 1$. Give an example with $b - a = 1$.

Q3-2001-25. Prove that the number of integer-sided triangles with perimeter n equals the number of partitions of n into parts 2, 3, 4 in which 3 appears at least once.

2002 — Twenty-Sixth Competition, IASBS of Zanjan

Q1-2002-26. Let $f : [a, b] \rightarrow [a, b]$ be continuous, differentiable on (a, b) , with $f(a) = a$ and $f(b) = b$. Prove there exist distinct $x_1, x_2 \in (a, b)$ with $f'(x_1)f'(x_2) = 1$.

Q2-2002-26. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $f(f(f(x))) = x$ for all x . Prove $f(x) = x$ for all x .

Q3-2002-26. Define $M(n, k)$ to be the number of minimal k -element covers of an n -element set. Prove

$$M(n, n - 1) = \frac{n}{2}(2^n - n - 1).$$

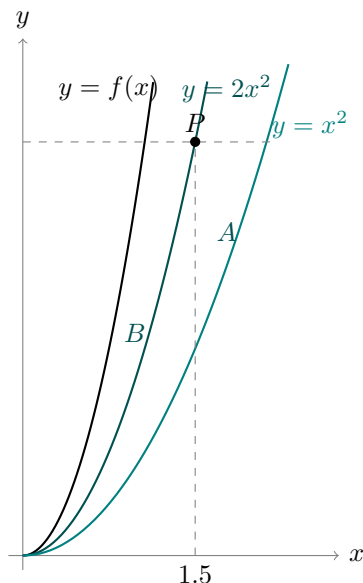
2003 — Twenty-Seventh Competition, Bu-Ali Sina University, Hamedan

Q1-2003-27. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable, $a < b$, with $f(a) = f(b) = 0$, $f'(a) > 0$, $f'(b) > 0$. Prove f' has at least two roots in (a, b) .

Q2-2003-27. Prove that $[0, 1]$ cannot be written as a disjoint union of closed intervals each of positive length strictly less than one.

2004 — Twenty-Eighth Competition, Sharif University of Technology

Q1-2004-28. Let f be one-to-one and continuous. For every point P on the curve $y = 2x^2$, the regions A and B shown in the figure have equal areas. Determine f explicitly.



Q2-2004-28. Let $f : [a, b] \rightarrow (a, b)$ be continuous. Prove that for every $n \in \mathbb{N}$, there exist $\alpha > 0$ and $c \in (a, b)$ with

$$f(c) + f(c + \alpha) + f(c + 2\alpha) + \cdots + f(c + n\alpha) = (n + 1) \left(c + \frac{n}{2} \alpha \right).$$

Q3-2004-28. Let A consist of all binary n -tuples having an odd number of 1's. Determine, with reasoning, the maximum number of vectors from A any two of which share an even number of 1-coordinates.

2005 — Twenty-Ninth Competition, University of Mazandaran

Q1-2005-29. Let $f : [0, a] \rightarrow \mathbb{R}$ be continuous and positive. Prove that

$$\left(\int_0^a f(x) dx \right) \left(\int_0^a \frac{dx}{f(x)} \right) \geq a^2.$$

Q2-2005-29. Consider the sequence 1, 11, 111, 1111, ... Prove that if $\gcd(m, 30) = 1$, then infinitely many terms are divisible by m .

2006 — Thirtieth Competition, Tafresh University

Q1-2006-30. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be continuous with $\lim_{x \rightarrow \infty} f(x) = 1$. Evaluate

$$\lim_{n \rightarrow \infty} \int_{1385}^{2006} f(nx) dx.$$

2007 — Thirty-First Competition, Ferdowsi University of Mashhad

Q1-2007-31. A circle is colored with two colors. Must there exist a monochromatic inscribed triangle which is (a) equilateral, (b) right-angled, (c) isosceles?

Q2-2007-31. Let n be an odd natural number. Prove there exist n consecutive natural numbers whose sum is a perfect square. Do there exist 12 consecutive natural numbers whose sum is a perfect square?

Q3-2007-31. Let $x_1, \dots, x_{2n}$ ($n \geq 2$) be reals such that removing any one of them, the remaining $2n - 1$ can be partitioned into two sets with equal sums. Prove that all x_i are zero.
